

# Option F Formula Rationale (short note)

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## Formula used in `Experiments-06-05-ABM-F.nlogox`

In the sanction update block (offender and punishment witnesses), misconduct propensity is updated by:

```

$$\Delta \text{misconduct} = - \text{learning-rate} * \text{punishment-severity} * ($$


$$\text{reporter-protection} * (1 + \text{punishment-severity}) / 2$$


$$- 0.8 * \text{punishment-severity} * (1 - \text{reporter-protection})$$


$$)$$

```

## Why this structure?

The formula combines two opposing mechanisms:

### 1. **Deterrence channel (negative term)**

`reporter-protection * (1 + punishment-severity) / 2`

- Stronger protection increases credible reporting and therefore expected sanction impact.
- Punishment enters again as a signal-amplifier  $((1 + P)/2)$ , so high sanctions can have stronger marginal deterrence.

### 2. **Backlash / fear-trap channel (positive counterforce inside bracket)**

`0.8 * punishment-severity * (1 - reporter-protection)`

- If protection is weak, harsher punishment can increase concealment pressure, perceived unfairness, or reactance.
- This weakens (or can locally reverse) deterrence, creating flat or upward segments in some mechanism curves.

## Main levers and what they do

- `learning-rate`: global speed of behavioral adaptation.
- `punishment-severity`:
  - increases deterrence,
  - but also increases backlash under low protection.
- `reporter-protection`:
  - strengthens deterrence,
  - suppresses backlash via  $(1 - \text{reporter-protection})$ .
- `0.8` (backlash coefficient):
  - calibration lever for nonlinearity and fear-trap strength,
  - higher values produce more flattening / upward bends at low protection.

## What this enables in plots

Compared with purely monotonic deterrence formulas, Option F can produce:

- mostly downward curves for medium/high protection,
- but flat or partially upward curves for low protection at high punishment,
- stronger nonlinearity in heatmaps (instead of near-linear gradients),
- clearer separation between "Iron Fist" and "Safe Harbor" policy archetypes.

## How to justify empirically

Use existing outputs (`exp1_heatmaps`, `exp1_mechanism`, `exp1_archetypes`, `exp2_pareto`) and report:

1. **Mechanism evidence:** Identify specific protection levels where slope changes from negative to near-zero/positive at high punishment.
2. **Fear-trap evidence:** Show policy cells with low hidden misconduct reduction but elevated fear/retaliation under low protection + high punishment.
3. **Archetype contrast:** Compare **Iron Fist** (high P, low R) vs **Safe Harbor** (low P, high R) on hidden misconduct and retaliation.
4. **Robustness:** Repeat with different seeds/replications and confirm the pattern is not noise.

## Conceptual references (for framing, not strict structural estimation)

- Near & Miceli: expected-utility logic in whistleblowing.
- Dozier & Miceli (1985): retaliation risk and reporting context.
- Mesmer-Magnus & Viswesvaran (2005): antecedents of whistleblowing/reporting behavior.
- Sherman (1993) Defiance Theory: sanctions can backfire under perceived illegitimacy/low fairness.